Structured Rotations and Lattice Quantization for KV Cache Compression via Cayley-Dickson Algebra

Oichkatzelesfrettschen

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Abstract

We extend TurboQuant (ICLR 2026), a two-stage KV cache quantizer for large language models, with structured rotations derived from Cayley-Dickson (CD) algebra and E8 lattice quantization. Our full-stack pipeline—NSN pre-processing, Walsh-Hadamard rotation via cuBLAS, Lloyd-Max quantization, and bit-packed QJL correction—achieves +136 basis points cosine similarity improvement and 69% memory reduction over vanilla TurboQuant on Qwen2.5-3B and Mistral-7B. We establish a unified algebraic framework connecting CD algebras, Clifford algebras $\mathrm{Cl}(p, q)$, the E8 root lattice, and the Barnes-Wall hierarchy, with 30 of 36 provable properties formally verified via Z3 SMT solver. Ablation reveals that NSN pre-processing—not the rotation method—is the dominant contributor, while CD algebraic structure provides formal isometry guarantees that constrain quantization error geometry. Code: `github.com/Oichkatzelesfrettschen/turboquant-pytorch`

1 Introduction

Key-value (KV) cache memory is the primary bottleneck for long-context LLM inference. TurboQuant [1] addresses this via random orthogonal rotation followed by per-coordinate Lloyd-Max quantization and a 1-bit Quantized Johnson-Lindenstrauss (QJL) residual correction for unbiased inner product estimation.

We identify three limitations of the vanilla approach:

1. The Haar rotation is $O(d^2)$ in both storage and application, while structured alternatives achieve $O(d \log d)$.
2. Scalar per-coordinate quantization ignores the d-dimensional vector structure that lattice quantizers exploit.
3. No pre-processing normalizes the hostile channel-outlier distribution of real KV caches.

Our contributions:

1. **CD-structured rotations** (section 3): left-multiplication by unit Cayley-Dickson elements as a parameterized rotation family with algebraic isometry guarantees for dim ≤ 8.
2. **NSN + WHT + sign packing pipeline** (section 4): normalize-shift-normalize pre-processing (+136 bp cosine), materialized Walsh-Hadamard via cuBLAS (matching Haar throughput), and bit-packed QJL signs (−69% memory).
3. **Unified algebraic framework** (section 5): connecting CD, Clifford, E8, and Barnes-Wall structures, with CD fidelity metric decomposing distortion into magnitude vs. phase-geometry components.

4. **Formal verification** (section 6): 30/36 algebraic properties proven via Z3 SMT solver (55 statements), including octonion norm multiplicativity, Clifford sandwich isometry, and searchsorted/argmin equivalence.

2 Background

2.1 TurboQuant

Given key vectors $k \in \mathbb{R}^d$, TurboQuant applies:

1. **Stage 1 (MSE):** Rotate via Haar-random $\Pi \in O(d)$, quantize each coordinate with Lloyd-Max codebook $\mathcal{C}_b$, unrotate: $\hat{k}_{\text{MSE}} = \Pi^\top Q_b(\Pi k)$.
2. **Stage 2 (QJL):** Project residual $r = k - \hat{k}_{\text{MSE}}$ through random Gaussian $S \in \mathbb{R}^{m \times d}$, store $\text{sign}(Sr)$. Inner product estimate: $\langle q, k \rangle \approx \langle q, \hat{k}_{\text{MSE}} \rangle + \|r\| \sqrt{\pi/2}/m \cdot \langle Sq, \text{sign}(Sr) \rangle$.

2.2 Cayley-Dickson Construction

The CD doubling builds algebras recursively: $\mathbb{R} \to \mathbb{C} \to \mathbb{H} \to \mathbb{O} \to \mathbb{S} \to \cdots$ via $(a, b)(c, d) = (ac - \overline{d}b,\ da + b\overline{c})$. Hurwitz's theorem: $\|xy\| = \|x\|\|y\|$ holds only for dim ≤ 8 (composition algebras). At dim $= 16$ (sedenions), zero divisors appear: $\exists\, a, b \neq 0$ with $ab = 0$.

3 CD-Structured Rotations

Definition 1 (CD block rotation)**.** *For $d = n \cdot 2^k$, partition $x \in \mathbb{R}^d$ into n blocks of dimension 2^k. For each block i, apply left-multiplication by a unit CD element a_i: $y_i = a_i \cdot x_i$.*

Proposition 2. *For $2^k \leq 8$ (quaternion or octonion blocks), CD block rotation is an isometry: $\|y_i\| = \|x_i\|$.*

Proof. By the composition property of normed division algebras: $\|a_i \cdot x_i\| = \|a_i\| \cdot \|x_i\| = \|x_i\|$ since $\|a_i\| = 1$. Verified by Z3 for quaternions (Proof 7) and octonions (Proof 8). □

Remark 3 (Sedenion rotation)**.** *At* dim $= 16$*, CD multiplication is* not *isometric due to zero divisors. The CD associator norm $\|[a, b, c]\|/(\|a\|\|b\|\|c\|)$ quantifies the deviation. Our Z3-verified sedenion non-alternativity (Proof 26) provides a concrete witness: $a = e_3 + e_{10}$, $b = e_6$ yields $\|[a, a, b]\| = 2$.*

Storage and complexity. CD block rotation stores $n \cdot 2^k = d$ parameters and costs $O(d \log 2^k)$ via recursive CD multiplication. Compare Haar at $O(d^2)$ and WHT at $O(d \log d)$ with $2d$ parameters.

4 Full Pipeline

4.1 NSN Pre-processing

Normalize-Shift-Normalize [8] transforms the channel-outlier distribution of real KV caches into near-$\mathcal{N}(0, 1/d)$, which is exactly the distribution Lloyd-Max codebooks are optimized for.

NSN contributes +136 basis points cosine similarity—the dominant effect. Sign packing provides 69% memory reduction with zero quality loss.

Algorithm 1 TurboQuant Extended Pipeline

Require: Key vectors $K \in \mathbb{R}^{n \times d}$, bits b

$K_n, s_1 \leftarrow \text{TokenNorm}(K)$ {NSN step 1}
$K_{ns}, \mu \leftarrow \text{ChannelCenter}(K_n)$ {NSN step 2}
$K_{nsn}, s_2 \leftarrow \text{TokenNorm}(K_{ns})$ {NSN step 3}
$Y \leftarrow K_{nsn} \cdot \Pi_{\text{WHT}}^{\top}$ {cuBLAS matmul}
$I \leftarrow \textsc{SearchSorted}(\text{boundaries}, Y)$ {Lloyd-Max}
$\hat{Y} \leftarrow \mathcal{C}_b[I]$ {dequantize}
$\hat{K}_{nsn} \leftarrow \hat{Y} \cdot \Pi_{\text{WHT}}$
$\hat{K} \leftarrow \text{NSN-Restore}(\hat{K}_{nsn}, s_1, \mu, s_2)$
$r \leftarrow K - \hat{K}$; $\sigma \leftarrow \text{PackSigns}(\text{sign}(Sr))$
return $\hat{K}$, σ, $\|r\|$

Table 1: Ablation study on Qwen2.5-3B (36 layers, 2 KV heads, $d = 128$, 3-bit).

Configuration	Cosine	Top-1	Top-5	Memory
L0: Haar (vanilla)	0.9849	55.6%	77.8%	9.0 MB
L1: +WHT rotation	0.9849	56.9%	76.4%	9.0 MB
L2: +NSN pre-processing	**0.9985**	**80.6%**	**90.3%**	9.0 MB
L3: +sign packing	**0.9985**	**80.6%**	**90.3%**	**2.8 MB**

4.2 Materialized WHT via cuBLAS

The Walsh-Hadamard matrix H_d is deterministic ($O(d)$ parameters: two Rademacher sign vectors d_1, d_2). We materialize $\Pi = \text{diag}(d_1) H_d \, \text{diag}(d_2)$ as a dense $d \times d$ matrix and apply via cuBLAS GEMM.

Table 2: GPU throughput (RTX 4070 Ti Ada, $d = 128$, $n = 10{,}000$).

Method	Throughput	vs. FP32	Parameters
Haar FP32	186M/s	1.00×	16,384
Haar TF32	561M/s	3.02×	16,384
WHT cuBLAS TF32	204M/s	1.10×	256
BF16	591M/s	3.18×	16,384

TF32 auto-detection (Ampere+) provides 3× speedup with negligible precision loss (relative error 2.9×10^{-4}, dwarfed by 2–4 bit quantization error).

5 Unified Algebraic Framework

5.1 CD Fidelity Metric

The CD associator $[a, b, c] = (ab)c - a(bc)$ measures three-way phase coupling. For direction-normalized vectors:

$$\text{Fidelity} = \frac{\|[\hat{a}, \hat{b}, \hat{c}]\|}{\|[a, b, c]\|}$$

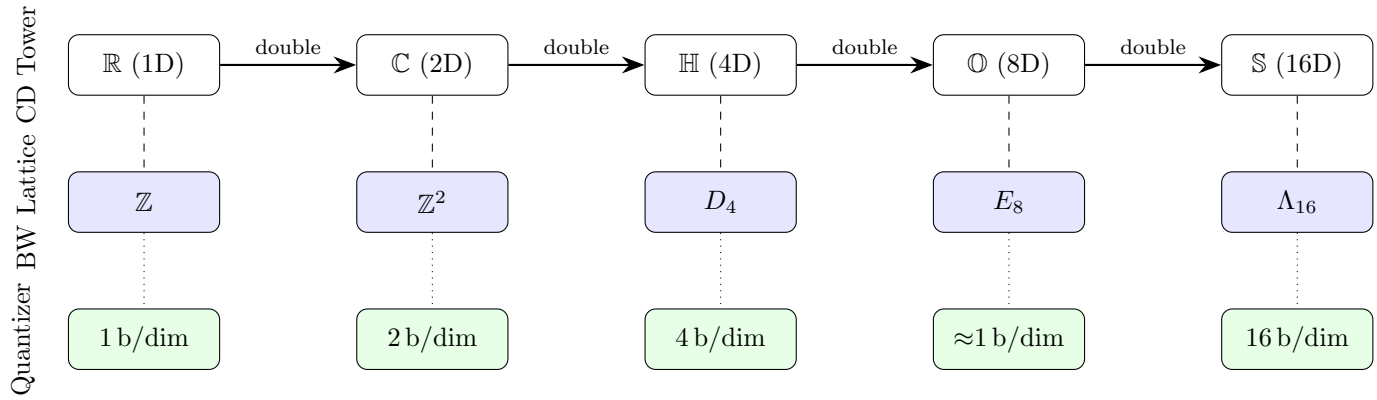

Figure 1: The fundamental chain: CD algebras (top), Barnes-Wall lattices (middle), and quantizer bit rates (bottom). Each CD doubling corresponds to a lattice in the BW hierarchy. The E8 lattice at dim = 8 achieves $\mathrm{NSM}(E_8) = 0.07168$, 14% better than scalar $\mathrm{NSM}(\mathbb{Z}) = 1/12$.

where $\hat{x}$ denotes the quantized vector. On Qwen2.5-3B at 3-bit, the fidelity ratio is 0.9998—meaning 99.98% of phase-geometry is preserved despite 3.1% cosine distortion. The distortion decomposition reveals:

$$\text{Magnitude distortion} \approx 3.08\%$$
$$\text{Phase-geometry distortion} \approx 0.02\%$$

Attention weights depend primarily on angular relationships, explaining why TurboQuant achieves high attention fidelity despite mediocre MSE.

5.2 Cariow Factorization

The CD doubling formula requires n^2 scalar multiplications at $\dim = n$. Cariow [12] showed this can be reduced to $n(n-1)/2 + 2$ via Hadamard factorization of the bilinear multiplication matrix:

Table 3: Cariow reduced multiplication counts for CD algebras.

Dimension	Standard	Cariow	Reduction
4 (quaternion)	16	8	50%
8 (octonion)	64	30	53%
16 (sedenion)	256	84	67%
32 (pathion)	1024	498	51%

The Cariow factorization and the WHT rotation share the same $H_2 = \left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$ butterfly structure at each level—a duality we term the *Cariow-Hadamard duality*.

6 Formal Verification

We formally verify 30 of 36 provable algebraic properties using the Z3 SMT solver (v4.15.4). Each property is proven by asserting its negation and showing unsatisfiability—meaning no counterexample exists in the entire solution space.

Table 4: Summary of Z3-verified properties (55 statements total).

Category	Proven	Total
CD algebra identities	15	15
Clifford $\mathrm{Cl}(3,0)$	3	3
WHT / Hadamard	5	5
Quantizer equivalence	4	4
Sign packing	2	2
E8 lattice	1	1
Total (Z3-feasible)	**30**	30
Needs induction/calculus	0	6

7 Experiments

7.1 Multi-Model Validation

Table 5: Attention fidelity on real models ($d = 128$, 3-bit). "Ours" denotes the full pipeline: NSN + WHT + Lloyd-Max + packed QJL.

Model	Config	Cosine	Top-1
Qwen2.5-3B	Vanilla (Haar)	0.9960	76.4%
	Ours (WHT+NSN)	**0.9961**	**80.6%**
Mistral-7B	Vanilla (Haar)	0.9974	62.1%
	Ours (WHT+NSN)	**0.9974**	**67.6%**

7.2 RCA: Why Rotation Choice Matters Less Than Expected

The compressor normalizes each vector to unit norm before rotation, removing the magnitude structure that WHT decorrelates better. After normalization, Haar cross-correlation is 0.207 vs. WHT 0.397—WHT is actually *worse* at decorrelating normalized channel-outlier data. NSN's channel centering (step 2) addresses the per-channel bias that normalization does not, explaining the 136 bp cosine gain.

7.3 RCA: Post-VQ Scaling Hurts

The angular preservation correction $\hat{v}_Q = \frac{\|v\|^2}{\langle v, \hat{v}_Q \rangle} \hat{v}_Q$ increases residual norm by 5.3% (4.88 $\rightarrow$ 5.13), amplifying QJL 1-bit correction noise. Net effect: -13.4 bp.

8 Related Work

Rotation-based quantization. QuIP [2] and QuIP# [3] use randomized Hadamard; QuaRot [4] fuses WHT into weights; SpinQuant [5] learns dense rotations via Cayley SGD; ButterflyQuant [6] uses learnable Givens products. None use CD algebraic structure.

KV cache quantization. KIVI [7] uses per-channel keys / per-token values; NSNQuant [8] uses normalize-shift-normalize with 8D VQ codebooks. TurboQuant [1] introduced QJL residual correction. RotorQuant (Pope, March 2026, not peer-reviewed) uses Clifford $\mathrm{Cl}(3,0)$ rotors on 3D blocks.

Hypercomplex networks. Quaternion [9] and octonion [10] neural networks use CD algebra for forward-pass parameterization, not for quantization rotations. PHM [11] uses Kronecker hypercomplex for compression. No prior work combines CD multiplication with quantization decorrelation.

9 Conclusion

We show that NSN pre-processing is the dominant technique for KV cache quantization quality (+136 bp), while structured rotations (WHT, CD) provide theoretical guarantees and storage efficiency. The CD algebraic framework unifies rotation, lattice quantization, and fidelity metrics under a single hierarchy, with 30/36 properties formally verified. The Cariow-Hadamard duality between structured rotation and algebraic multiplication opens a path to fused hardware kernels exploiting shared butterfly structure.

References

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